Exam Prep AI - Mock Question Paper

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Reference Paper

03UCEST105122402

ALGORITHMIC THINKING WITH PYTHON

Max. Marks: 60

Duration: 2 hours 30 minutes

PART A

(Answer all questions. Each question carries 3 marks)

1. Write an algorithm to check whether a given string is a rotation of another given string.

CO-1 (3)

2. Design a flowchart to find the sum of all digits in a given integer.

CO-2 (3)

3. Write a Python program to generate the first 'n' numbers in the Fibonacci sequence using a divide and conquer approach.

CO-3 (3)

4. Devise a greedy algorithm to find the most efficient way to pack items of different sizes into a box of fixed size.

CO-2 (3)

5. Write a recursive function to find the number of ways to climb 'n' stairs in 'k' steps, where each step can be either 1 or 2.

CO-3 (3)

6. Write an algorithm to check whether a given matrix is symmetric or not.

CO-1 (3)

7. Write a Python program to simulate a population increase using a random walk model.

CO-3 (3)

PART B

(Answer any one full question from each module, each question carries 9 marks)

Module -1

9. Explain how backtracking strategy can be applied to solve the 0-1 Knapsack problem.

CO-1 (5)

10. Your college is located in a metropolitan city and you are new to that city. You would like to go out for dinner with your friends. Explain how heuristics approach can be used to find the best restaurant for dinner.

CO-1 (4)

Module -2

11. Write a pseudocode to implement a grading system based on the following conditions.

• Marks ≥ 90: Grade A

• Marks 80–89: Grade B

• Marks 70–79: Grade C

• Marks 60-70: Grade D

• Marks 50-69: Grade E

• Marks < 50 : Grade F

• If the difference between the marks and the next multiple of 10 is less than 3, then convert it to the next grade.

• If the marks are below 50, no need to convert even if the difference is less than 3.

CO-2 (6)

12. You are a software developer working on a data analysis tool for a logistics company. The company has a list of delivery times, where each entry represents the time taken (in minutes) for a particular delivery. The team needs to identify delivery times that are divisible by 6 (as they are considered optimal delivery durations) but not divisible by 4 (as they tend to involve more complexity in scheduling). Write an algorithm to print the delivery times that are divisible by 6 but not divisible by 4 from a given set of delivery times.

CO-2 (3)

Module -3

13. You are working as a software engineer for a financial analysis company. The company uses Fibonacci sequences to model certain financial growth patterns, such as project milestones or revenue projections. You are tasked with developing a tool that generates the first ‘n’ numbers in the Fibonacci sequence to assist in creating projections for clients. You should use recursion to solve it.

CO-3 (5)

14. Write a program that accepts the length of three sides of a triangle as input and determines whether or not the triangle is a right triangle.

CO-3 (4)

Module -4

15. You are given two positive integers, a and b. Write a Python program using recursion to find the Least Common Multiple (LCM) of these integers. You are not allowed to directly calculate the LCM formula, but must instead use the relationship between GCD and LCM.

CO-3 (6)

16. You are working on a text-processing application for a content moderation team. The team needs a feature to analyze user-generated sentences and reorganize their characters for better readability and analysis. Specifically, they want all capital letters to appear first, followed by small letters, then digits, and finally any special characters. Write a Python program to implement this feature, ensuring the reorganized output is displayed as a single word.

CO-3 (3)